# SAMIYA KAZI

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## EDUCATION

**Vidyalankar Institute of Technology, Mumbai** 2023 – 2027
B.Tech, Electronics & Telecommunication | **CGPA 9.8/10** | Minors: Computer Science; Next Generation Data Science
**Jai Hind College, Churchgate** — HSC (Class XII), Science 2023
**St. Agnes' High School, Mumbai** — SSC (Class X) 2021

## EXPERIENCE

**Data Analyst Intern — Vanguard Ventures, Mumbai** May – Jul 2026

- Cleaned a property walk-in tracker in pandas, fixing a broken header, a corrupted formula column and ~700 swapped dates.
- Defined 10 sales-funnel KPIs with explicit denominators: Channel Partners drove 77.2% of walk-ins, and a misleading "96% qualified" figure turned out to be missing data (only 6.6% of leads were ever tagged).
- Built a self-updating Excel dashboard and a web dashboard; recommended CP incentives and mandatory lead tagging.

**Digital Marketing Intern** — Reincarnation NGO Mar – Apr 2025
**Crowdfunding Intern** — Muskurahat Foundation (raised ₹5,000) Jan – Feb 2024

## PROJECTS

**FloodWatch — flood forecasting & satellite mapping for Assam | Live site | GitHub**
*Python | LightGBM | PyTorch U-Net | Sentinel-1 SAR | Time-series | GitHub Actions | D3.js*

- Forecast floods 1–7 days ahead at 12 Brahmaputra-basin stations from river flow and rainfall; walk-forward test on six monsoons (2021–26) caught 90% of flood days a day ahead (CSI 0.82 vs 0.73 baseline).
- Validated against ASDMA district flood reports: impact model beat river-level and seasonal baselines (CSI 0.41 vs 0.32); rainfall nearly quadrupled 3-day early warnings in the 2022 floods (11% → 42%) at a 0.9% false-alarm rate.
- Trained a U-Net on Sentinel-1 radar to map floodwater through cloud (0.65 water IoU vs 0.23 for Otsu); daily live forecast via GitHub Actions, with leakage tests in CI.

**LitScout — AI research agent for verified literature reviews | Live app | GitHub**
*Python | LLM agents | Tool calling | Groq (gpt-oss, Qwen) | OpenAlex & arXiv APIs | TF-IDF ranking*

- Built a public research agent: anyone types a topic and it plans searches, screens hundreds of papers, follows citation trails and writes a review; a verifier checks every cited sentence against its source.
- Benchmarked against the reference lists of 7 published surveys across fields: citation snowballing found 6× more expert-cited papers than keyword search (14.6% vs 2.3% recall@200); zero invented citations in generated reviews.
- Deployed as a public web service (FastAPI, Docker); per-task model routing, fallback and caching keep it on free tiers.

**Zomato Real-Time Lakehouse — streaming big data platform | Live demo | GitHub**
*Kafka | Spark Structured Streaming | Delta Lake | Airflow | dbt | PostgreSQL | Grafana | Spark MLlib | FastAPI | Docker*

- Built an end-to-end lakehouse for a simulated food-delivery marketplace: Kafka → Spark streaming into bronze/silver/gold Delta tables (watermarks, de-duplication, quarantine) → Postgres, run by 5 Airflow DAGs with 15 dbt tests.
- Trained Spark MLlib late-delivery models (gradient-boosted trees, ROC-AUC 0.71, 69% of late orders caught) that score every live order in seconds, plus K-means customer segments; 11-tab analytics site with a guarded, read-only LLM text-to-SQL assistant.

**AquaSense AI — IoT water-quality & microplastic monitoring** Team of 4
*ESP32 | YOLO | FastAPI | SQLite | Flask*

- ESP32 sensor node (pH, TDS, turbidity, temperature) with a YOLO detector classifying microplastic polymers from UV-lit ESP32-CAM images; FastAPI backend and role-based portal with live map and analytics; research paper in preparation.

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, C/C++, Java, JavaScript, MATLAB
**Machine Learning & Data Science:** scikit-learn, XGBoost, LightGBM, Random Forest, anomaly detection (Isolation Forest), time-series forecasting, walk-forward validation, feature engineering, Pandas, NumPy, EDA, Matplotlib
**Deep Learning & Computer Vision:** PyTorch, TensorFlow/Keras, CNNs, transfer learning (MobileNetV2), U-Net segmentation, YOLO, OpenCV, Tesseract OCR, satellite SAR imagery
**Generative AI & NLP:** LLMs, AI agents, tool calling, RAG / Self-RAG, LangChain, LangGraph, embeddings, Sentence Transformers, FAISS, semantic search, prompt engineering, LLM evaluation, text-to-SQL
**Big Data & Engineering:** Apache Spark (PySpark, Structured Streaming, MLlib), Kafka, Delta Lake, Airflow, dbt, PostgreSQL, SQLite
**Deployment & Tools:** FastAPI, Flask, React, Streamlit, Docker, Git/GitHub, GitHub Actions, Excel, Power BI, D3.js, Jupyter
**Cloud:** Google Cloud (Cloud Run, Cloud Storage); cloud computing fundamentals (C-DAC certified)

## CERTIFICATIONS & ACTIVITIES

Cloud Computing (C-DAC) · MATLAB Onramp · Class Representative (60+ students) · Creatives team, VIT & Jai Hind · Badminton, basketball